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December 19th, 2021

Foucault and Information Technologies: The Owl of Minerva

The owl of Minerva fascinates in the most salient way. Hegel's belief that philosophical understanding of historical conditions only manifest as that condition fades from existence reminds me of Plato's paradox of inquiry. If we knew what the historical condition was why would we bother with inquiry to its effects? If we do not know what the historical condition is then how is it feasible to pragmatically contextualize the current condition? The historical condition we ought to understand subsequently will only appear as that condition is reflected upon, and thereby already passed.

Michael Foucault's paradigms of discipline and surveillance (panopticism), in this respect, are a philosophical understanding of political relations (specifically human – human) in the past. Not to the degree wherein all aspects of Foucauldian theories should be dismissed, yet to a degree which is not encompassed by those theories. With the introduction of algorithms to daily life we ought to consider a paradigm shift of discipline and surveillance. Contextualizing the change in Foucault's theory will require analysis of his fundamental understanding of discipline. Therein, we can understand how it has changed juxtaposed to other presented theories of how algorithms effect discipline and surveillance.

Following the summation of Foucault, I contend his theory with Shoshana Zuboff's article, *Big Other: surveillance capitalism and the prospects of an information civilization*, to demonstrate what aspects of Foucauldian theory I believe will still be relevant. Zuboff argues information being mediated by algorithms 'automates' and 'informs' social relations. She concludes these two effects textualize, and thereby politicizes, the social relation wherein

computer-mediated algorithms are used. I found Zuboff's 'alternative' theory similar to Colin Koopman's in his book *How We Become our Data*. Koopman argues the fastening and formatting nature of algorithms represents 'infopower.' Koopman also states it is similar to biopower, discipline, and sovereign power but distinguishes that infopower can define social relations that the others cannot.

Firstly, I believe it is noteworthy to focus on the foundation of Foucault's paradigms of discipline and surveillance to understand where our current reality diverts from Foucauldian theories understanding of our historical 'condition.'

In Michel Foucault's *Discipline and Punish: The Birth of the Prison*, he writes, in "the body of the condemned," of the ontological effect of the exercise of power on the condemned man.¹ Foucault argues the doubling of the condemned man is "That of a 'non-corporal', a 'soul,'" which acts as a "technology of power over the body" of itself. This noncorporeal-body relation "is born... out of methods of punishment, supervision, and constraint" and after becomes a tool for perpetuating the current power relations; thereby reinforcing domination and altering behavior with discipline.²

These factors are important. Understanding in what ways discipline, surveillance, and constraint has changed in social, and thereby political, relations will illuminate whether Foucault's theory can account for algorithmic domination of the subject. To define the extent Foucault's 'soul' influences the individual he writes,

The man described for us, whom we are invited to free, is already in himself the effect of subjection much more profound than himself. A 'soul' inhabits him and brings him to existence, which is itself a factor in the mastery that

¹ "Condemned man" referring to Foucault's thought experiment; inverting the role of the Monarch, who experiences a doubling of his body.

² Foucault, Michel, 1926-1984. "The body of the condemned," *Discipline and Punish: The Birth of the Prison*, New York: Pantheon Books, 1977. p 29.

power exercises over the body. The soul is the effect and instrument of a political anatomy; the soul is the prison of the body.³

For the condemned man the structural design of imprisonment is designed in such a way to dominate—through discipline, surveillance, and constraints—the subject’s ‘soul’ over its own body. It is not necessarily that the subjected man willingly submits himself to domination over his very being. Algorithms, I believe, uses this internalized noncorporeal power to a different and more unlimited degree.

Within Foucauldian theory this noncorporeal technology of power is informed by the vector of power that is the social relation between the prison and prisoner; the very “materiality” of the panopticon, and most prisons, uses the ‘soul’ as “one of its tools” to punish and discipline. Within this framework the condemned man is subjected by his own soul through *overt* ‘methods of punishment, supervision, and constraints.’⁴

Algorithms, I believe, uses this internalized noncorporeal power to a different and more unlimited degree. Unlimited in the sense that technology has no limit to what aspects of life it becomes intertwined with. Furthermore, algorithmic discipline, surveillance, and constraint on individuals differs from the manner of the power relations of the panopticon.

Firstly, discipline, in Foucault’s use, is a method of punishment wherein a human-to-human relation is one of dominator and dominated. This relation must be adapted to include algorithms as actants within their own right. I do not believe algorithms ‘knowingly’ do what they do, but algorithms are simultaneously a cause and effect of ontological change.

Secondly, Foucauldian supervision is one of an interesting relation. The capacity to be surveilled is overt; in the sense that the subjected man is aware of the possibility of being

³ Ibid. Foucault. p 30.

⁴ Ibid. Foucault. p 30.

monitored. The understanding of the possibility is the overt nature of surveillance. The actual monitor, though, is in a sense hidden from the condemned man, out of sight in the panopticon's tower in complete anonymity.

Algorithmic relations present an interesting diversion from this dual sense of surveilling actants containing anonymity and no anonymity. I believe the relation has inverted. The condemned man now has complete anonymity to his counterparts, the other prisoners. In social relations an example of this would be a new, 'unformatted' Facebook profile. Ironically, the anonymity of the user is willingly given up. This is because the user is simultaneously under complete anonymity with other users while having no anonymity with arbiter of the algorithm.

For example, even though the aforementioned Facebook profile has no information *publicly* shared, the company (viz. its algorithm) collects location data, login data, and all behavior done through the algorithm. How this role of anonymity has changed, I believe, is where Foucauldian theory lacks the most pragmatic applications to algorithmic relations.

Thirdly, the authority of power relations with constraint as a tool of exercising power have undoubtedly shifted towards a noncorporeal sense of identity. For example, Foucault describes every prison revolt and the "innumerable discourses that prison has given rise to since the early nineteenth century have been about the body and material things." Constraints on the condemned man focus on domination over the physical body (the 'soul' as a tool).⁵

Algorithmic relations and the power they exercise, though, dispatches from this notion of constraint over a *physical* subject towards constraint on their digitized self. The 'soul' I believe still forms in algorithmic relations, but the role discipline, surveillance, and constraint has

⁵ Ibid. Foucault. p 30.

undoubtedly changed. Zuboff attempts to answer how these have changed with her ‘surveillance capitalism.’

Shoshana Zuboff argues ‘big data’ is “an autonomous process” which “is above all the foundational component in a deeply intentional and high consequential new logic of accumulation” which she terms surveillance capitalism. This new aspect of surveillance demonstrates how Foucault only accounts for the *effects* of surveillance. Insofar as ‘surveillance capitalism’ represents a dual edge; the surveillance imposes and informs; is cause and effect.⁶

Zuboff contends surveillance capitalism is encompassed by “computer-mediated work.” He argues there is a “fundamental duality that information has in formatting relations” and internalizing those formats as parts of the self. For example, the automation of this formatting is reflected in the increasingly determinant power market rational has in determining behavior.⁷ This aspect of Zuboff’s theory reflects the complexity of ‘constraint’ within algorithmic relations. In so far as Zuboff argues “[Computer-mediated work] not only imposes information (in the form of programmed instructions), but it also produces information” and thereby differs from the very one-directional nature of Foucault’s constraint on the condemned body.⁸

He further writes,

The word I coined to describe this unique capacity is ‘informate.’ Information technology alone has the capacity to ‘automate’ and to ‘informate.’ As a result of the informatting process, computer-mediated work extends organizational codification resulting in a comprehensive ‘textualization’ of the work environment – what I called ‘the electronic-text.’⁹

⁶ Shoshana Zuboff, “Big Other: surveillance capitalism and the prospects of an information civilization,” *Journal of Information Technology* 30 (2015), p 75.

⁷ ‘market rational’ refers to rational dictated by modes of market (cost-benefit-esque).

⁸ Ibid. Zuboff, p 75; ‘one-directional’ nature of constraint refers to the constraint on the prisoner’s body having has no capacity to impose directions and simultaneously produce new directions.

⁹ Ibid Zuboff, p 76.

These ‘unique’ capacities entail power relations that Foucauldian theories fall short of explaining discipline, surveillance, and constraint of individuals in technological social relations. In Foucauldian framework discipline, surveillance, and constraint act to manifest a ‘soul.’ In the panopticon the ‘soul’ is the ‘technology of power’ used in social relations to dominate the condemned man with his own soul. The goal being desired ontology or behavior.

In a sense Foucault’s surveillance has the purpose of mimicking the ‘soul.’ Because the soul has complete mastery over the body, being ‘the prison of the body,’ it follows that its nature is in line with that of the watchtower.

The ‘soul’ is constantly present in any behavior, intentionality, or thought. Nevertheless, that same ‘soul’ is noncorporeal; thereby physically nonexistent. Therefore, the surveillance of the panopticon must have embodied subjects insofar as embodiment is required to exercise power over those bodies. This ‘embodiment’ is transitioning towards a digital medium wherein surveillance is neither overt viewership nor hidden eyes.

Zuboff describes this change as the ‘textualization’ of the work environment. Wendy Brown’s *Undoing the Demos*, in “Political Rationality and Governance,” offers an extension to Foucauldian framework. She describes Foucault’s neoliberalism as an “economic rationality” that “becomes a governing (or political) one” creating new power relations, “but also governing subjects themselves and every institution on the landscape” of social relations (i.e., schools, army, hospitals, etc.).¹⁰

Wherein Zuboff’s textualization follows this trend of imposing an economic rational; inherent in ‘big data’ is utilizing the efficient nature of algorithms to not only sift through that

¹⁰ Wendy Brown. 2015. “Undoing Democracy: Neoliberalism’s Remaking of State and Subject,” *Undoing the Demos: Neoliberalism’s Stealth Revolution*. p 121.

data but to collect more. However, asserting Foucault's modes of markets governing social and political relations applies within algorithmic relations seems intuitively incorrect.

It is not necessarily that social, and thereby political, relations with algorithms cannot be dictated by modes of market, but that this does not capture the complete relationship between humans and 'electronic text.' I propose the aspect Foucault will always be missing is spatial in nature.

In other words, the exercise of power in Foucault's understanding is necessarily a consequence of corporeal domination between one physical body over another in relation to each other (albeit using a noncorporeal vector of power to enforce said domination).

Zuboff argues, to this point, that "electronic text and what can be learned" from those textualized environments will never "and can never be— 'things in themselves'" for "they are already embedded in the social, their possibilities circumscribed by authority and power." Thus, it is not that surveillance has necessarily changed *who* it dominates, but *how* it does so. For example, textualization of the work environment dominates users because the algorithmic relation is inherently already a power structure. Those being dominated are being informed and automated.¹¹

While the panopticon necessarily required the corporeal relation between the watchman and the watched, surveillance capitalism encompasses a transition towards informatting and automating all social relations in an algorithmic way. For example, Foucault's condemned man subjects himself because of physical anguish and ontological belief of being watched by a physical object.

¹¹ Ibid. Zuboff, p 77.

Now imagine a Starbucks wherein there are cameras pointed in every direction. The register itself has an algorithm to keep track of the balances. Online orders are filtered through Starbucks servers, and these inform the employees. Within each order, though, the surveillance has changed from being of a *corporeal* body to a noncorporeal identity. The cameras throughout the store do not matter. What matters is the relation noncorporeal subject on the Starbucks servers have.

This noncorporeal identity is masked in the textualization within apps. For example, when filling out a mobile order for Starbucks you are asked to sign in. ‘Signing in’ requires not only for the individual to attribute traits (like name, birthdate, etc.) but also Starbucks preferences (favorite location, drinks, foods).

Foucault cannot account for this changed dynamic. The user’s online activity in this example is inevitably tracked and recorded (by account of it being digital). However, the exercise of power is no longer over watching that corporeal person but watching their noncorporeal digital activity. Informatting and automation, thereby, are forces which transition the subject away from embodiment towards a ‘logic of accumulation.’

Though, Zuboff argues this transition towards a logic of accumulation is not historically unique. An example is the transition from corporate capitalism (mass-production) to financial capitalism over the 20th century. It is this pervasive nature of computer mediation which will alter how “every aspect of the world is rendered” through “new symbolic dimensions as events, objects, processes, and people become visible, knowable, and shareable in a new way.” What Zuboff describes here is what Foucault is missing; the relations and ontological effects between fellow users has utterly changed in the last fifty years.¹²

¹² Ibid. Zuboff, p 77.

Zuboff's theory of electronic text does not appear complete enough to capture these new relations, though. For example, anonymity, and the lack thereof, are effects of relations between a human and algorithm. The informing and automation of the algorithm manifests or destroys that anonymity. For example, a user of Facebook has complete anonymity until they do not. The traits that define the online profile are algorithmic vectors of power which informate and automate the user. Yet this does not capture the psychological effects of that profile being in relation to *billions* of others.

Not only is the profile in relation to other users, but the information in the profile is defined by characteristic and traits, inherently noncorporeal, *chosen* or *omitted* by the user. The profile being released is an informatted and automated noncorporeal identity of a corporeal being. The profile is what the user wants others to see. Foucault might contend this new relation between a digitized identity and the existent 'soul' reflects the same power exercised over the condemned man. I believe, though, this identity is an entirely new extension of the self.

A user's extended identity is still dominated by Foucault's 'soul.' This 'soul' is informed by 'electronic text' (informatting and automated) and thus is having power exercised over its noncorporeal self. I believe Colin Koopman explains well why Foucauldian framework of biopower and discipline cannot necessarily capture the existence of two identities mentioned above (the condemned man and the digital user).

Koopman's framework is encompassed by his coined term 'infopower.' Koopman argues his "claim is not that infopower replaces or negates other modes of power —" biopower, sovereign power, disciplinary power— "only that it operates beyond" the limits of those modes.

This is exemplified in the unknown ontological effects of users around the world being available to share and to be known by every other user.¹³

What distinguishes ‘infopower’ from other vectors of power aforementioned is its fastening and formatting capabilities. Curiously similar to Zuboff, Koopman offers a framework of how information technology influences the subject (simultaneously being effect and cause of power dynamics). With regards to defining ‘infopower,’ Koopman writes, infopower

... is a distinctive mode of power; its ‘formatting’ refers to a general array of *techniques* on which this power relies; its ‘fascinating’ refers to the way in which power is *operated* via these techniques in a dual project of both ‘canalizing’ and ‘accelerating’ operations; and its ‘informational persons’ are the subjects, or targets, of this mode of power... Information itself is today a formidable mode of rationality by which [knowledge is gained].¹⁴

Koopman notes the importance of seeing information technology as a mode of rational. He argues the data is justified because the data it came from imposes such a notion. Furthermore, that imposed notion then produces more information which fastens to the user.

The user, with the rational of infopower, fastens themselves to traits, or data ‘canalizing’ users towards noncorporeal attributes (formatting viz. recommendation). These attributes are then justified by the very data which formats the user to specific traits (height, eye color, interests, etc.). The result of such dynamics is the production of data, and justification for that production, from “data itself” by which users format their identities, and thereby, fasten themselves to algorithmic data.¹⁵

¹³ Colin Koopman, *How We Became Our Data: A Genealogy of the Informational Person* (Chicago: University of Chicago Press, 2019), p 163.

¹⁴ Ibid. Koopman, p 159.

¹⁵ Ibid. Koopman, p 159-160.

A great example Koopman uses to illustrate such a relation is the “politics of racialization at the level of printed blank forms with their little check boxes” next to the traits. With regard to Koopman’s framework, formatting the user to specific characteristics or personalities through a racial lens is justified by the fastening effects of ‘blank forms.’ These forms “enacted a politics that fastened people to their data” to the extent wherein “Persons who previously racialized by a different set of techniques began to become fastened to their race through informational technologies of forms.”¹⁶

Formatting and fastening, though, are simply Koopman’s answer to how Foucault’s biopower and genealogical forces do not extend into the realm of informational technologies. The formatting and fastening itself seems as if a technique to further distance Foucault’s ‘soul’ from the subject’s embodiment within that social relation. In other words, as these techniques, fastening and formatting (similar to Zuboff’s textualization viz. informatting and automizing), are done so to systematically categorize users wherein any individual subject can be identified by the information fastened to them.

For example, imagine Foucault’s condemned man. The condemned man is at the whim of power relations insofar as they constrain his physical self. Although constraint is not always physically coercive; the power exercised to constrain the condemned man is undoubtedly a corporeal relation, yet the effects are noncorporeal.

In Zuboff and Koopman’s framework these relations have shifted from a corporeal cause (wherein constraint was on the subject) towards a digitized, noncorporeal self. I think one ought to believe Foucault’s ‘soul’ is still present within the subject, but it is no longer the ‘prison of the

¹⁶ Ibid. Koopman, p 165.

body.’ In fact, constraint and surveillance have shifted drastically away from corporeal forces to a power simultaneously informing and fastening users to perceived noncorporeal characteristics.

An important thing to note is the informing and fastening of characteristics, personalities, traits, etc. to a user are not to discipline us on “how we should act or how we must act.” In fact, it is much more sinister and covert. While fastening is “focused more on forms” formatting is “focused on delineating the shapes that we already are” in such a way that who we are is, in part, informed by the categorizations offered by algorithms. The algorithm is simply giving multiple choices and users are picking those which they associate to themselves.¹⁷

The key takeaway is embodiment is transitioning towards its opposite wherein fastened traits are offered and defined by information technology; thus, we become defined by such relations. How algorithms “pin us rather to our traits, which are construed as anything but physical and material” is an example of such a transition away from embodiment.¹⁸

Clearly Foucault’s theories are not as accurate as they once were. In part, I believe, because of the rise of automation and informatting being what dictates our relations to algorithms. Drone warfare, for example, fastens traits to targets. These traits, being noncorporeal, are prioritized for surveillance; insofar as fastening traits to be prioritized by a drone algorithm supposedly is done because it optimizes the rate at which enemies of the state are found. However, this optimization is as such that individuals who also pertain certain noncorporeal traits might be fastened to that prioritized agenda. In this sense, the discipline, surveillance, and constraint are now to a more extreme degree than before.

The panopticon is now the world. The surveillance is in the plethora of political and social relations wherein information technologies both informate, automate, and fasten traits to

¹⁷ Ibid. Koopman, p 166.

¹⁸ Ibid. Koopman, p 166.

individuals. Drones using facial recognition, for example, fastens traits unto subjects, which are then justified by how much the data matches (data justifying data). However accurate this informational technology may become, there will always be a chance the individual, even with a 99% match, is not the desired target. This is because what is being formatted and imposed are noncorporeal traits which an algorithm fastens unto a *real* being.

Furthermore, the power exercised by information technology is similar to capacities of a drone insofar as power can be operated upon bodies remotely across the world. The discipline and force dominating Foucault's condemned man was overt and required direct relation to the subject. Information technology has shifted from this dynamic by giving individuals the capacity to become available to all through algorithms.

Instagram, for example, possess an interesting formatting trait called 'verified.' The idea is that if you are publicly published you receive a blue check saying, 'this user is who she says she is.' Somehow the capacity to be *known* by many begets a formatting which reassures other users this noncorporeal information (because Instagram is 1's and 0's on a database) is a *real* person. The oxymoron of it is delightful.

Furthermore, I feel the largest part lacking in Foucault's framework is the psychoanalytic effect of such a format (verified sticker) on all of users. Not only does it demonstrate a difference between technological selves, but it is also constantly puts users in a hierarchical social relation to each other. Because individuals are *more known*, 'individuals' referring to their data online, they are formatted and fastened in such a way that their sense of self is more complete.

To be honest, I am at a loss for what exactly I mean by this 'completeness.' It is not that the formats available to me are not capable of define all that I perceive myself as. But insofar as this completeness is a part of information technology that is gatekept. My intuitive decision to

use this term comes from the inversed nature of information technology compared to Foucault's discipline, surveillance, and constraint. However, I would not be surprised if these inversed relations will not be fully understood until all of humanity has transferred to a digitized version of ourselves. Once we have completed shifted away from the corporeal domain of political relations a philosophical understanding will come about. Maybe this is what Foucault meant endeavoring, quite oxymoronicly, to write "the history of the present."¹⁹

¹⁹ Ibid. Foucault, p 31.